DESIGN AND EVALUATION OF A SMART BIN MONITORING AND INCENTIVIZED WASTE MANAGEMENT INSTITUTIONAL PILOT MODEL

A project synopsis submitted in partial fulfillment of the requirements for the award of the degree of

Master of Computer Applications

by

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1. ABSTRACT

In Manipur, the collection, segregation and disposal of wastes is an area which needs massive work. Currently, the people of Manipur rely heavily on the Imphal Municipal Council(IMC) and several other NGOs for the collection and disposal of solid wastes in Imphal, Manipur. Imphal West and East alone with a combined population of about 3 lakhs, generates approximately 110 metric tons of solid waste daily. And the IMC and NGOs collect and dispose only about 50 percent of the wastes generated daily. A loophole in public-private partnerships have been reported with regular complaints from the public households due to irregular collection of waste. As it stands, the public just waits for their household wastes to be collected which as stated can be sometimes ineffective. The IMC and NGOs follow a strict schedule for their collection from their respective wards as well which means if a bin overflows or a household wants to rid their waste, they have to wait almost a week or dump it themselves somewhere. In view of these local challenges, the proposed project develops a low-cost smart waste-management infrastructure that links real-time bin monitoring with collection task generation, worker coordination and user participation. Smart bins equipped with fill-level sensors will transmit readings to a backend system. When a configurable threshold is reached, the system will automatically generate a collection task and notify the responsible collection personnel. An administrator dashboard will provide centralized visibility of bin status, task progress, collection history and response performance. A user module will support QR-based disposal tracking, issue reporting and incentive points for responsible disposal and segregation. The system will initially be evaluated as an institutional pilot in Imphal, providing a controlled environment in which technical reliability and operational usefulness can be measured before wider deployment is considered. The evaluation will consider sensor reliability, notification latency, collection response time, task completion, overflow incidents, usability and user participation against the conventional workflow. The expected contribution is a localized, affordable and scalable computational framework that can complement existing municipal and NGO collection arrangements in Imphal.

2. OBJECTIVES

The primary objectives of this research project are:

- To examine the existing waste collection, segregation and disposal workflow relevant to institutional settings in Imphal.
- To identify operational gaps such as delayed collection, inadequate monitoring, irregular task allocation and limited source segregation.
- To design and develop a low-cost IoT-enabled system for real-time monitoring of waste-bin fill levels.
- To implement automated collection-task generation, worker notifications and task-completion tracking.
- To provide QR-based disposal tracking, issue reporting and incentive-based participation for responsible waste segregation.
- To evaluate the proposed system using technical, operational, usability and behavioural measures and assess its potential for future public-private partnerships in Imphal.

3. LITERATURE SURVEY

The proposed work bridges the existing gaps by integrating monitoring, collection task allocation, user engagement, and measurable evaluation within a low-cost institutional framework.

Table 1: Literature Survey Comparison

Author / Ref	**Method / Focus**	**Key findings / Advantages**	**Limitations / Gap**
Zoumpoulis et al. [1]	Systematic review of smart bins	Maps sensing and automation trends	Standardization and maturity challenges
Mahongnao [2]	Study of urban solid waste management in Imphal, including collection, transportation, disposal, segregation and policy.	Identifies continued reliance on conventional collection and disposal, weak source segregation, infrastructure needs and the importance of public participation and PPP.	Policy/management-oriented; does not develop a computational mechanism for monitoring or coordinating collection.
Greater Imphal Master Plan [3]	Official planning assessment of solid waste management in Greater Imphal under the revised GIS-based master plan.	Provides planning context for waste generation, collection, treatment, source segregation and infrastructure requirements.	Planning-level document; does not implement or experimentally evaluate an IoT task-management solution.
Wang et al. [4]	IoT and college waste behaviour	Links usability and behaviour	Does not focus on collection tasking
Proposed Work	**IoT fill-level monitoring + automatic task generation + administrator/worker application + user participation.**	**Integrates detection, task creation, collection execution, monitoring and user engagement in one localized workflow.**	**Requires prototype development and empirical evaluation in an institutional setting in Imphal.**

4. PROPOSED METHODOLOGY

The proposed framework consists of the following key phases:

- **Problem Analysis and Requirements:** Study the institutional collection workflow, identify stakeholders, and define functional, non-functional, and baseline requirements.
- **IoT Data Acquisition:** Use an ESP32-class controller and fill-level sensor to collect bin readings. Threshold confirmation/averaging will be used to reduce false alerts.
- **System Architecture / Application Design:** Develop the backend/database, administrator dashboard, worker task interface, and user module. Threshold events create collection tasks and notifications.
- **User Engagement and Evaluation:** Use QR-based disposal tracking and points for responsible disposal/segregation; evaluate sensor reliability, notification latency, response time, overflow incidents, task completion, monitoring effort, usability, and participation.

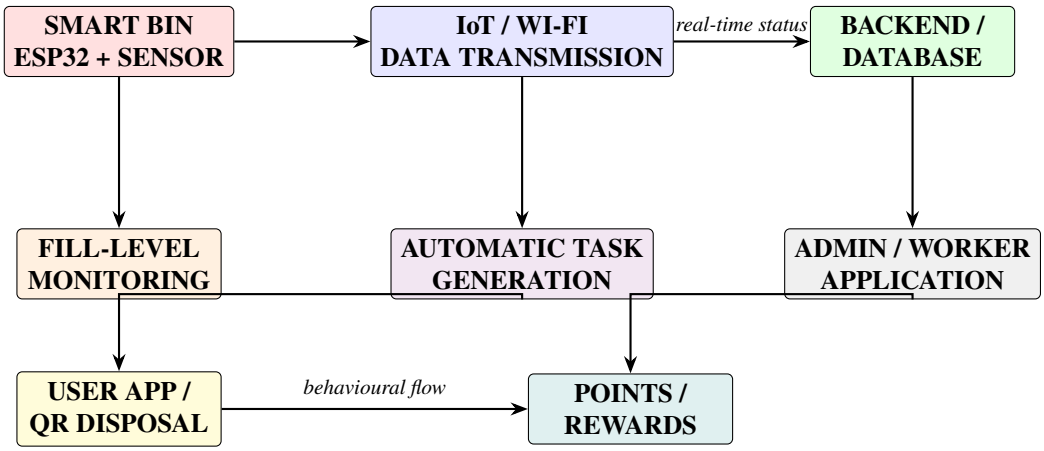


Figure 1: Proposed System Architecture and Data Flow Pipeline

As illustrated in Figure 1, sensor readings flow from the smart bin through the communication layer to the backend and database. Threshold events generate operational tasks for collection staff, while the user flow records disposal activity and incentives. The administrator dashboard provides centralized status and performance information.

5. MOST IMPORTANT CONTRIBUTION

- **Localized Problem-Solution Alignment:** Directly targets documented collection, segregation, infrastructure and participation gaps identified in Imphal.
- **Low-Cost, Context-Aware Design:** Uses affordable hardware and modular software with an institution-first deployment model suitable for resource-constrained environments.
- **Empirical Validation and Scalability:** Measures technical, operational, usability and behavioural outcomes while defining a pathway toward wider community and municipal integration.
- **End-to-End System Design:** Integrates bin monitoring, automatic task creation, worker execution and user interaction within a single workflow.

6. CONCLUSION

This synopsis proposes a smart institutional waste-management system designed around the documented realities of Imphal, Manipur. Local studies have identified persistent challenges in conventional waste collection and disposal, source segregation, infrastructure, public participation and waste processing. The proposed system addresses the technological and operational side of this gap by connecting IoT-based fill-level monitoring with automated collection-task generation, worker task tracking and user participation. The institutional pilot provides a controlled setting in which sensor reliability, notification latency, response time, task completion, overflow incidents, usability and participation can be evaluated before wider deployment is considered. The research is therefore intended to contribute a localized and affordable computational framework that complements existing municipal and NGO collection arrangements in Imphal.The people no longer have to wait on unreliable collection but rather can participate themselves while still being rewarded for it. This means anyone who is willing can take on the task of collecting waste themselves which makes it a form of employment. Future work may investigate multi-institution deployment, route-aware collection, broader municipal integration and an authorized community collection-partner model.

References

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[2] Mirinchonme Mahongnao, "Urban Solid Waste Management in Imphal, Manipur," *J*ournal of Industrial Pollution Control, vol. 33, no. 1, pp. 696–701, 2017.

[3] Department of Town Planning, Government of Manipur, "Revised GIS Based Master Plan for Greater Imphal – 2043 (1st Modification)," Solid Waste Management, Section 10.5, Final Gazette Notification, 2024.

[4] Y. Wang et al., "Understanding waste sorting behavior and key influencing factors through internet of things: Evidence from college student community," *Resources, Conservation and Recycling*, vol. 174, 105775, 2021. doi:10.1016/j.resconrec.2021.105775.